Examiner only

Answer **all** *questions.*

1. *(a)* The following diagrams show the life cycle of two different mass stars.
Diagram 1 shows how a star that is similar in mass to our Sun changes with time.
Diagram 2 shows the changes for a star that is 8 times more massive than our Sun.

Complete the labelling on diagrams **1** and **2**, using words or phrases from the box below. Words or phrases can be used once, more than once or not at all. [4]

asteroid	neutron star	red giant	supergiant	white dwarf

Diagram 1 – Star of mass similar to our Sun

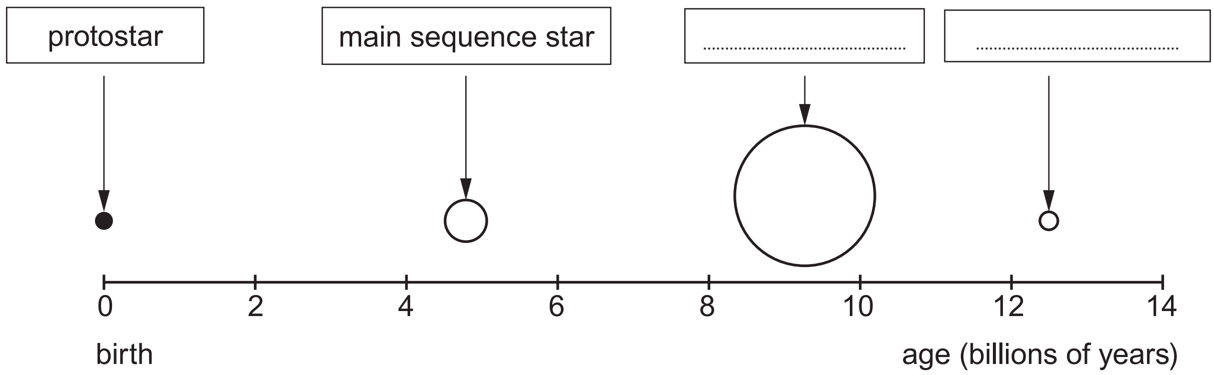
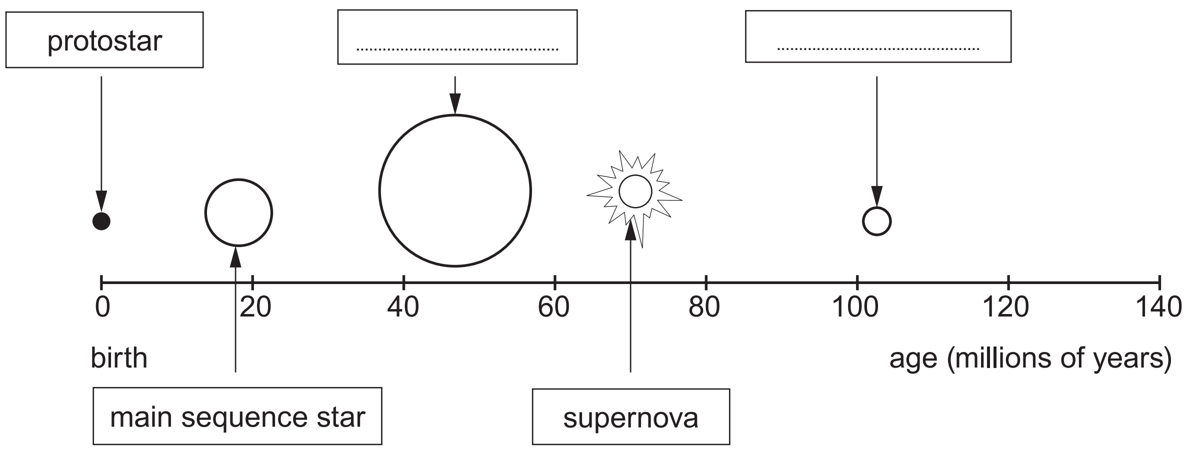


Diagram 2 – Star that is 8 times more massive than our Sun



 (3430U60-1)

Examiner only

(b) **<u>Underline</u>** the word or phrase in the brackets which correctly completes each sentence. [4]

(i) A main sequence star is stable because its gravitational force is (**less than** / **equal to** / **greater than**) the force caused by its radiation pressure.

(ii) Stars generate their energy by the (**burning** / **fusion** / **fission**) of increasingly heavier elements.

(iii) During the final stages in the life cycle of some stars heavy elements are ejected when the star becomes a (**gaseous giant planet** / **supernova** / **supergiant**). The collapse of a cloud of gas and dust combined with these heavy elements can eventually form a (**Solar System** / **galaxy** / **Universe**).

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